Q1:

- **(a)** Peak position shifts to lower wavelength
- **(b)** Peak width increases
- **(c)** Peak intensity decreases

Q2:

(a)

$$\delta t \cdot \delta E \approx \hbar$$

$$\delta t \approx \frac{h}{2\pi \cdot \delta E} \approx \frac{h}{2\pi \cdot h \cdot \delta \nu} \approx \frac{1}{2\pi \times 100 \times 10^6\ \mathrm{s^{-1}}}$$

$$\delta t \approx 1.59 \times 10^{-9}\ \mathrm{s}$$

(b)

$$\delta t \approx \frac{h}{2\pi \cdot h \cdot c \cdot \delta \bar{\nu}} \approx \frac{1}{2\pi \times 2.997 \times 10^{10}\ \mathrm{cm \cdot s^{-1}} \times 10\ \mathrm{cm^{-1}}}$$

$$\delta t \approx 5.31 \times 10^{-13}\ \mathrm{s}$$

Q3:

$$A = \epsilon c l \qquad A = -\log T \qquad T = \frac{I}{I_0}$$

$$A = 227\ \mathrm{L \cdot mol^{-1} \cdot cm^{-1}} \times 0.00252\ \mathrm{mol \cdot L^{-1}} \times 0.2\ \mathrm{cm}$$

$$A = 0.114$$

$$-\log_{10} \frac{I}{I_0} = 0.114$$

$$\frac{I}{I_0} = 10^{-0.114} = 0.769\ (76.9\%)$$

$$\%\ \text{Reduction in intensity} = 100 - 76.9 = \mathbf{23.1}\%$$

Q4:

$$\%T = 100 - 48.3 = 51.7\%$$

$$A = -\log_{10} T = -\log_{10} 0.517 = 0.286$$

$$c = \frac{A}{\epsilon l} = \frac{0.286}{423 \times 0.65} = 0.00104 \text{ mol} \cdot \text{L}^{-1}$$

$$C \text{ (in mg/L)} = 0.00104 \times 200 \times 1000 = \mathbf{208} \text{ mg/L}$$

Q5:

$$\frac{N_f}{N_i} = e^{-\Delta E/k_B T}$$

$$\frac{N_f}{N_i} = 0.001\% = 0.00001 = e^{-\Delta E/k_B T}$$

$$\ln\left(\frac{N_f}{N_i}\right) = -\frac{\Delta E}{k_B T}$$

$$-11.51 = -\frac{\Delta E}{1.381 \times 10^{-23} \text{ J} \cdot \text{K}^{-1} \times 500 \text{ K}}$$

$$\Delta E = 11.51 \times 1.381 \times 10^{-23} \times 500 \text{ J}$$

$$\Delta E = 7.947 \times 10^{-20} \text{ J} = \frac{hc}{\lambda}$$

$$\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{7.947 \times 10^{-20}} = 2.5 \times 10^{-6} \text{ m}$$

$$\lambda = \mathbf{2500} \text{ nm}$$

Q6:

- **(a)** Electronic transitions
- **(b)** Vibrational transitions
- **(c)** Rotational transitions
- **(d)** Nuclear spin transitions